



Applying PBL methodologies to the chemical engineering courses: Unit operations and modeling and simulation, using a joint course project

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ABSTRACT

Unit Operations and Modelling and Simulation have been for long time a staple of the academic formation of any chemical engineer. Both are integral to the analysis of any chemical process, discretizing it into smaller specific processes that can then be characterized and modelled through the solution of balance, thermodynamics and transport equations. However, students usually perceive these subjects as separate fields of knowledge, and they do not develop the ability to correlate and integrate them to solve real-world problems in their future profession. On account of this, a Project-Based Learning (PBL) methodology was proposed in the redesign of the abovementioned undergraduate courses, focusing initially in Unit Operations. This PBL method was implemented alongside a joint course project, consisting on the design, assembly and characterization of a centrifugal pump, to be analyzed experimentally and computationally. To assess the success of this methodology, a survey was conducted on the students after they finished their courses. The results were mostly positive (85%), as the students appreciated the design component of the project, considering that it benefits their learning process, as well as the challenge it presented. This difficulty forced them to resort to different sources of information and areas of knowledge, alien to those provided in the courses. The limitations of the chosen project revolved around its limited scope and lack of connection with other topics of the courses (i.e. distillation columns). These limitations will be addressed with the design of transversal projects, which can cover more of the subjects seen in both classes.

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1. Introduction

Chemical Engineering (ChEn) started in the 1880s, both in the United States and Europe, simultaneously. With the emergence of ChEn, John Hinchley in London (UK) and William Walker in Boston, Massachusetts (US) coined the term “Unit Operations” (UnOps), as a method of engineering analysis in the design of plant for the process industries. In hindsight, it was an important element in the differentiation of ChEn from other disciplines (Cohen, 1996). UnOps are specific processes (usually involving physical, rather than chemical changes), common throughout the chemical industry (McCabe et al., 2005). Examples of UnOps are heating, cooling, storage and transport of fluids, distillation, crystallization, filtration, pulverization and so forth. Hence, the design of complete

chemical plants could be divided into the design of linked pieces of equipment, whose performance could be analyzed and studied separately. The power of UnOps was enhanced gradually over the following decades by applied research, which was carried out in academic institutions and in the industry.

Nowadays, the use of computing and information technology has increased so exponentially that it plays an important role on the chemical process industry for design and optimization. Consequently, the subject of Modelling and Simulations (SIM) is a requirement on most undergraduate ChEn curricula. The objectives of this kind of course focus on three main aspects: *i.* to provide fundamentals of process modelling and simulation, *ii.* to give an appropriate computer-based experience to the students, and *iii.* to show different numerical methods and computational techniques available to solve ChEn problems (Upreti, 2017). Examples of the engineering problems analyzed in this course are: mass, energy, and momentum balance and transport; thermodynamics; chemical equilibrium and reaction kinetics; and modelling of chemical process equipment such as reactors, distillation columns, mixers, heat

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Table 1
Chemical Engineering curriculum at the Universidad de Los Andes (Accreditation ABET until 2024).

Semester	Courses						
1	Spanish requirement	Unianandes' Basic Cycle Course	Chemistry	Introduction to Chemical Engineering	Algorithmics and Object Oriented Programming	Differential Calculus	
2	Applications of Chemistry	Physics I	Experimental Physics I	Linear Algebra I	Integral Calculus with Differential Equations	Thermodynamics	
3	Probability and Statistics I	Chemical and Phase Equilibrium	Physics II	Experimental Physics II	Fundamentals of Chemical Processes	Vectorial Calculus	
4	Investment Decision Analysis	Unianandes' Basic Cycle Course	Experimental Design in ChEn	Fundamentals of Chemical Analysis	Transport Phenomena I	Cell Biology	
5	Elective in Science and Math	Fundamentals of Engineering: Elective	Constitution and democracy	Unianandes' Basic Cycle Course	Transport Phenomena II	Chemical Reaction Engineering	Requirement of Reading in English
6	Unrestricted elective course	Unianandes' Basic Cycle Course	Mid-Career Project	Separation Processes	Unit Operations	Introduction to Modeling and Simulation	
7	ChEn elective	Elective, Other engineering program	Unianandes' Basic Cycle Course	Unianandes' Basic Cycle Course	Analysis of Control Systems	Process Plant Design	Undergraduate Project Seminar
8	Unrestricted elective course	ChEn elective	ChEn elective	Optimization of Chemical Processes	Undergraduate Graduation Project	Foreign Language Requirement	

exchangers and turbomachines (i.e. pumps). It is important to highlight that a blend of different subject areas on the ChEn curriculums is used as background to SIM to solve complex problems, which clearly present challenges for conventional teaching methods.

At the Universidad de Los Andes, both UnOps and SIM, are junior level courses, taught in the second half of third year of the bachelor, along with another highly intensive course: Separation Processes (marked in bold in Table 1). Before they can take these courses, the students have to learn about fluid mechanics; heat and mass transport; reaction kinetics; and programming. These topics are covered in the courses underlined in Table 1. This occurrence has created a sense of aversion among students towards this semester, because of the combined workload that the three subjects require.

Each course has a total assigned academic load of 144 h, during 16 weeks, which means that every student requires at least three hours of class attendance, and six hours of independent study per week (for these particular courses, UnOps has three lab practices through the semester and SIM has 1.5 h per week of computer lab). However, previous student screenings, which scrutinized about academic load in the junior year, diagnosed academically overstressed students, with around 49 h of work above the expected load, per subject. This problem was mainly due to the high amounts of different tasks and time-consuming activities that all courses required, and the additional pressure caused on the students by having to arrange their limited time between all the different classes they were taking.

Attempting to alleviate this burden on the students, different approaches were proposed to reduce the academic load, while trying to improve the learning experience of the students of both, UnOps and SIM courses. As a pilot screening, UnOps was selected as the first course to be redesigned. The proposed methodology included two major changes in the structure of the course, which were: implementation of flipped classroom and a PBL teaching methodology, which focused on a joint course project with SIM. This study focusses on the latter one. The changes reduced the work load by around 25%, bringing it down to the expected total of 144 h, or around 9 h per week.

1.1. Project-based learning

Project-Based Learning, known as PBL, focuses on the idea that concepts and ideas should not just be memorized and applied in theoretical singular problems, but that students gain a deeper

understanding through the application these concepts in real-world situations, allowing them to relate these concepts with other fields of knowledge and even practical and actual problems of his/her future profession. In 2013, Mills y Treagust stated six key reasons why most engineering education programs are deficient in addressing the concerns of the modern society (Mills and Treagust, 2003):

- Most of the engineering curricula are focused on engineering sciences and technical courses, leaving aside the integration of these practical cases.
- Current programs do not provide enough design experience to students.
- Graduates lack communication skills and teamwork experience.
- Programs need to develop more awareness amongst students of the social, environmental, economic and legal issues.
- Faculty does not have practical experience.
- Teaching strategies needs to become more student-centered.

PBL has been adopted by many engineering programs as an alternative to alleviate the aforementioned concerns. In PBL, five learning moments can be identified (Blumenfeld et al., 1991; Krajcik et al., 1994, 2003): *i.* The students start with a driving question. This part is related with problem to be solved. *ii.* Once the problem has been identified, students brainstorm ideas based on concepts in the discipline. *iii.* Possible solutions are debated, engaging collaborative activities, in contrast with expert problem solving from traditional learning methodologies. *iv.* While the debate takes place, students deepen their knowledge by participating in activities normally beyond their ability, and *v.* The students create tangible products that address the driving question.

The solution proposed here attempted not only to reduce the workload of the students, by having a collaborative project between the two courses instead of individual project. It also reinforced the idea that all of the subjects that the students learn are not separate fields of knowledge, but that they are part of a whole and should be used conjunctively to solve engineering problems. Many ideas were brought up to solve this problem and to present the students with the concept that the two subjects could be combined, so they could understand them as a whole. Looking at the two courses, it was noticed that they can be easily connected with three topics, namely: heat exchangers, mixing, and turbomachinery. All of them allowed the integration between two of the learning objectives

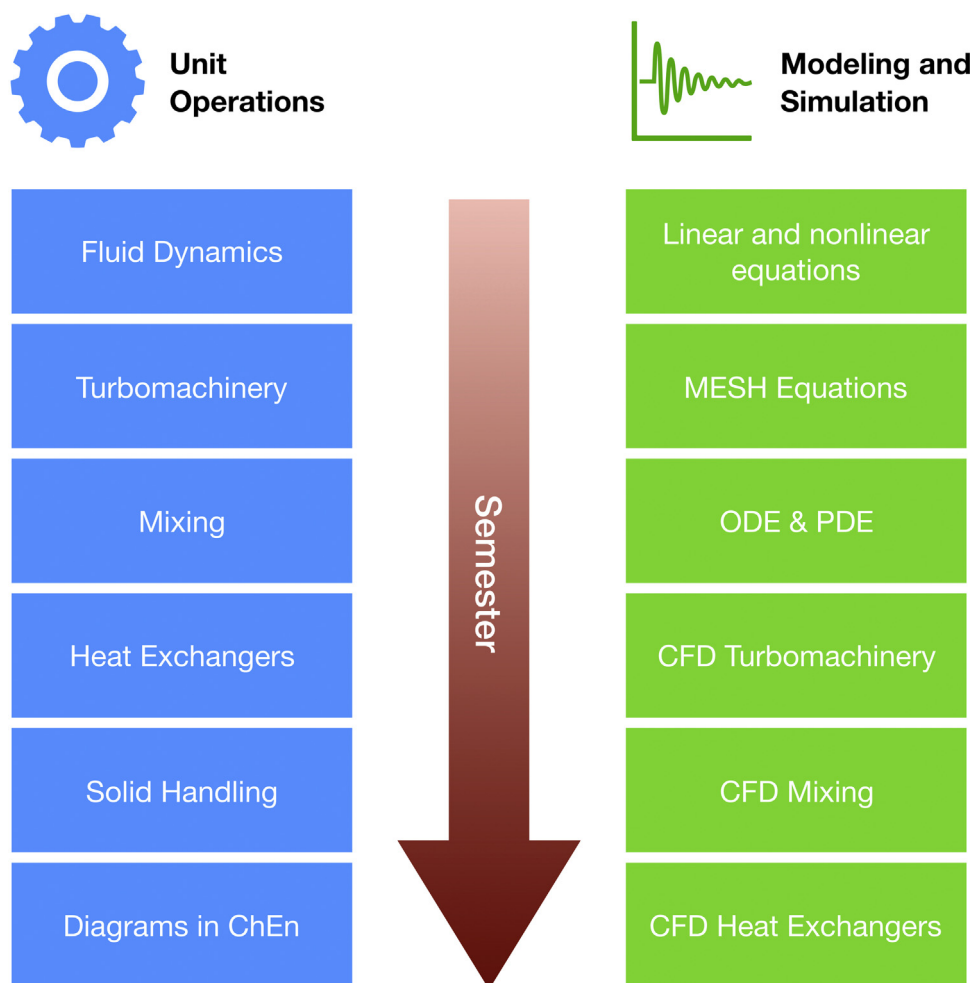


Fig. 1. Comparison between the Unit Operations and Modeling and Simulation courses program.

of the courses: *i.* to evaluate, establish dimensions and integrate unit operations in order to select the best one for a given process (UnOps), and *ii.* to apply tools and computational methods for the numerical solution of chemical engineering models (SIM). It should be noted that the learning objectives were not changed during the restructuring of the course.

From these three topics, the latter one, *turbomachinery* was selected as the core of the joint course project. The design and operation of a centrifugal pump was the specific topic chosen, and it is a common equipment in most operating plants. This decision was mostly arbitrary, but a couple main reasons were taken into account for this selection. First, fluid mechanics and transportation are one of the first topics taught during the semester, both in UnOps and in the CFD section of SIM (Fig. 1). This allowed the project to be commenced quite early in the semester, thus giving the students enough time to perform all the project steps. Second, this topic allowed a good balance between three aspects: *i.* an interesting design challenge, *ii.* a straightforward comparison criterion between designed products, and, *iii.* an accessible and feasible construction with the available lab resources.

The design of a centrifugal pump requires both the design of the casing and the impeller, with mostly just restrictions on sizing. They can also be compared with easily measured and reliable parameters, such as pump head and efficiency. A mixing system design, on the other hand, would most likely be restricted to the impeller, since the tank geometry would have to be the same for the systems to be comparable. On the other hand, most operational parameters would be either relatively stochastic (like residence or

mixing time) or difficult to measure (such as the blade shear rate). A centrifugal pump can also be printed with relative ease using polymer blends, while the construction of a heat exchanger would require the use of conducting and insulating materials, which may not be so easily shaped into the desired design. With this in mind, the design, construction, simulation and experimentation of a centrifugal pump was set as the joint project topic. UnOps oversaw the theoretical and experimental parts and SIM the computational modelling, simulation and validation (Fig. 2). The objective of this study was then to assess whether the joint project could improve interconnected learning without increasing work load on students or academic staff.

These activities had the aim to make the students connect abstract ideas to concrete applications to enable effective and cumulative learning as it is the purpose of the Legitimation Code Theory (LCT) (Auret and Wolff, 2017). In many of the courses there is a lack of connecting the theory with the application of it in the industry. That was one of the main goals of the project, to let the students identify how the study of a system will affect its main parameters and how these parameters will affect in a real situation in the industry. The LCT is an idea developed by the work of Bernstein and Bordieu and social realism, in which the ideas of semantic gravity (concerning the degree of connection between meanings and the contexts in which they find application) and semantic density (concerning complexity in terms of the condensation of meanings within practices) are discussed (Blackie, 2014). These two concepts are represented in Fig. 3.

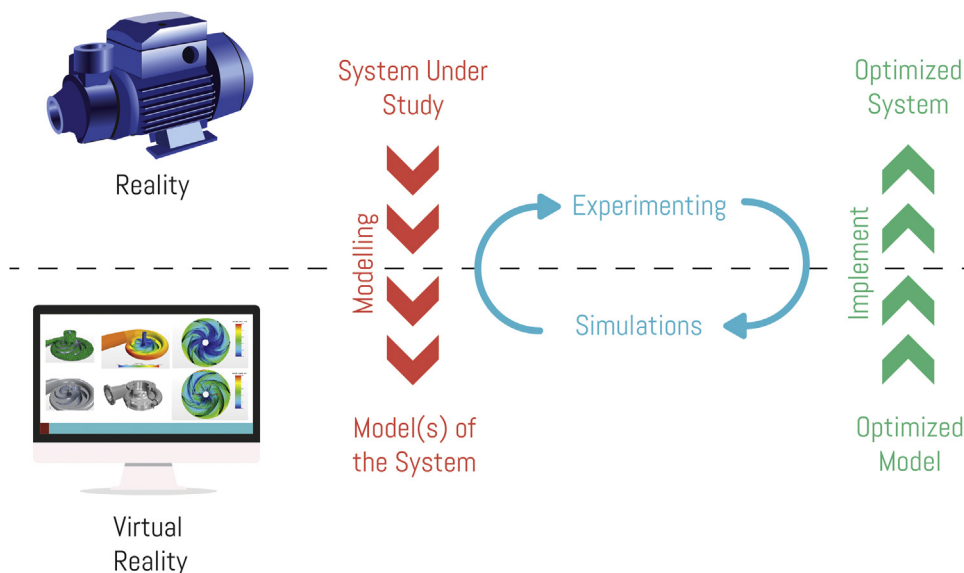


Fig. 2. Joint course project interactions between UnOps and SIM.

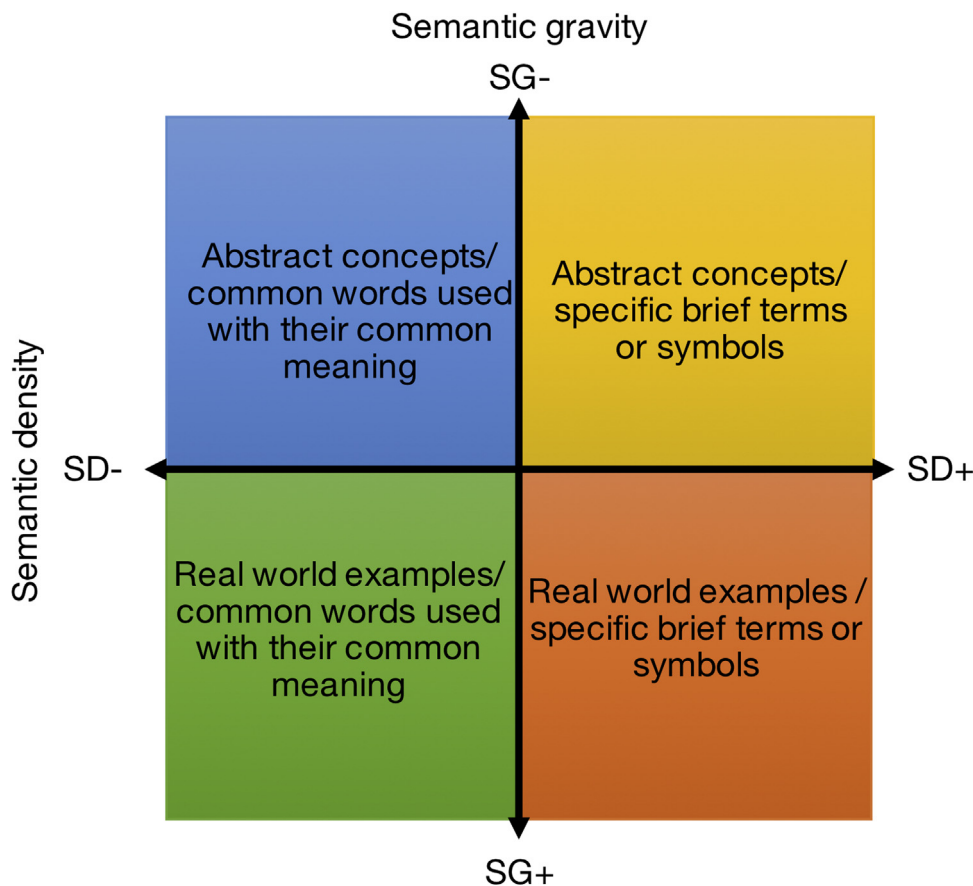


Fig. 3. Relationship between the two semantics as described by (Blackie, 2014).

2. Methodology

The main objective of the project was to design, build and characterize a centrifugal pump computationally and experimentally, determining its operating curve and efficiency. The designed pump had to have an innovative design, generating the largest head during its range of operation based on some constraints (i.e. the size of the pump had to fit in a cube of $12 \times 12 \times 12$ cm and

the rotation speed had to be 750 rpm). The quality of deliverable product (3D printed pump) was evaluated by comparing their performance of all proposed designs created by the different student teams.

The project was divided into three phases as shown in Fig. 4: i. Design of the geometry based on literature review and theoretical approximations, drawn in the software Autodesk Inventor® (Autodesk, Inc, USA), ii. 3D Printing of the new design; using a

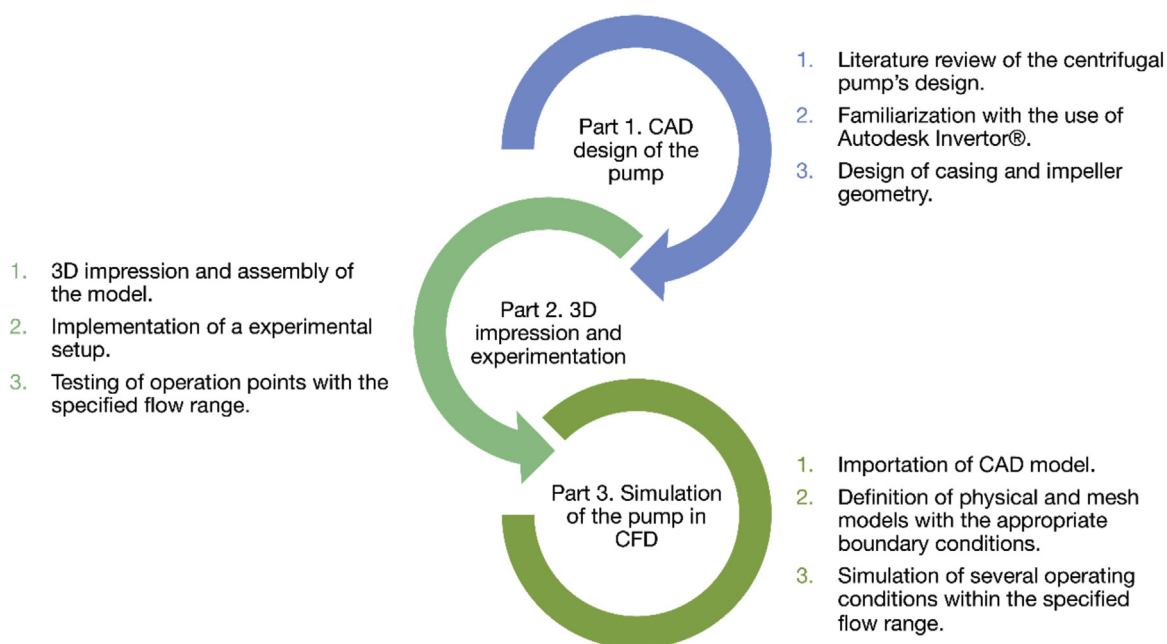


Fig. 4. Three-parts-based methodology.

Stratasys Mojo3D® printer and a thermoplastic polymer composed of Acrylonitrile, Butadiene and Styrene (ABS); followed by assembly and experimental data collection, and *iii*. Importation of the geometry to the Computational Fluid Dynamics (CFD) program STAR-CCM+v12 (Siemens AG, Germany), simulation of pump operation, and comparison with the results obtained experimentally. The last two phases were done complementarily. The two courses were planned so the same topics were studied at similar times.

In order to achieve the project's objective, from the UnOps point of view, the students needed first to envision a geometry for the pump (housing and impeller) considering their knowledge of the inner workings and requirements of a centrifugal turbomachine. To aid them in the design and sizing process, they were recommended Gülich's design guidelines (Gülich, 2010), where the author summarized the main equations used to correlate geometric and operating parameters. Then, they had to recreate that geometry in a computer-aided design (CAD), so they could 3D print it and assemble it. A first revision of their design was done at this stage, before the printing, to ensure that the pieces fitted and could be assembled together. They had to implement an experimental setup to study the operation of the equipment, determine the operational curve that relates the generated head with the impulse flow, and calculate the hydraulic power, the mechanic power and the efficiency curve with analyzed flows.

Likewise, from the SIM perspective, they had to import the proposed geometry to the CFD program, implementing the physics and mesh models with the boundary conditions that represented the system. It was left to the students to decide which models they had to use to better represent the system, granted they had to simulate the same geometry printed for the experimental setup, with water at atmospheric conditions used as operating fluid. The theoretical background behind these physics and mesh models had already been instructed to the students before this step of the project. Here-with, they had to construct the operation, power and efficiency curves, running several simulations at various flow rates, and validate the simulations against the experimental data. Finally, they had to compare the results with literature and evaluate if they were appropriate. If the behavior of the pump deviated too much from

the expected values, or an error was noticed after the printing stage, the groups were given one more chance to improve the design and print it once more.

In order to quantitatively measure the overall opinions of the students towards the proposed joint course project, and its structure as a PBL, a student online survey was conducted at the end of the semester taking into account seven main aspects: *i. Project Preparation*, *ii. Learning Strategies*, *iii. Background*, *iv. Project Description*, *v. Cognitive Challenge*, *vi. Cooperation*, and *vii. Review and Evaluation*. Each main aspect contained a series of questions, where students could grade it with more detail (Table 2). The students had to grade the validity of each statement with a whole number between 0 and 4. Additionally, a simple Yes/Not Sure/No question was asked at the end about whether the students considered that the joint course project benefited their learning process. For each question, the arithmetic mean of the answers of all students was then calculated. The survey was based on the work of Fernando Trujillo Sáez (2014) from the institute Conecta13, of the University of Granada.

3. Results and discussion

As mentioned earlier, the joint project aimed for students designing a centrifugal pump using the CAD software and testing its performance using two methods: experimental tests and CFD modeling. At the end of the course, students were able to assemble their own designs using 3D printing, estimating the corresponding characteristic curves of the centrifugal pump experimentally and computationally (by means of CFD) based on the knowledge acquired in both courses (Fig. 5). Additionally, several aspects of the pump's operation were thoroughly analyzed in both environments, such as recirculation or dead zones, pressure drop, and cavitation phenomena, allowing the students to apply the theoretical background obtained in both courses in a real-world situation. Moreover, the proposed joint project allowed the students to create a tangible product in order to address the driving question proposed, which was related to the factors affecting the operation of a centrifugal pump (i.e. head, power, efficiency and cavitation).

Table 2

Proposed survey to analyze the proposed joint project.

Project preparation	Indicators of project success were clearly defined for the evaluation. There is a clear relationship between the activities and expected competences of the students. The project is the result of the integration of different areas of knowledge. Significant activities are proposed for different capabilities, levels and learning styles of the students. The goals and contents are according departmental curricula. The learning goals are clearly defined. The final product is described, and it is related with the goals, contents and evaluation criteria.
Background	New knowledge is linked to previous courses and their own context. Relationships are established between previous and new knowledge.
Project description	All the steps to be followed (including temporal sequence) are detailed. The procedure for the presentation of the final product of the project is clear. The deadlines are clearly marked and reasonable.
Cognitive challenge	The goals pursued with the development of the project are clearly explained. Students are required to solve a problem of adequate complexity. Students are required to find information and assess their. Students are required to make connections between various sources of information and assess their suitability to solve the project. Students are required to perform activities in their surroundings for the resolution of the project. People, other than the teacher, (i.e. teaching assistants, manufacturers, etc.) are required to contribute their knowledge for the resolution of the project.
Learning strategies	Sufficient opportunities are given for students to use different learning strategies. Scaffolding techniques are used to help and support students (e.g. experiments and demonstrations). A variety of techniques are used to clarify the concepts (e.g. class examples, audiovisual aids and analogies).
Cooperation	Sufficient materials are used to make the project understandable and meaningful. The project has a cooperative structure. Interdependence and individual responsibility within the project are sought. Elements of co-evaluation are included.
Review and evaluation	A complete review of the fundamental knowledge in the development of the project is made. A response about student production is regularly provided. Formative evaluation moments are contemplated in which the student can make changes based on the feedback. A variety of assessment strategies are used throughout the task.

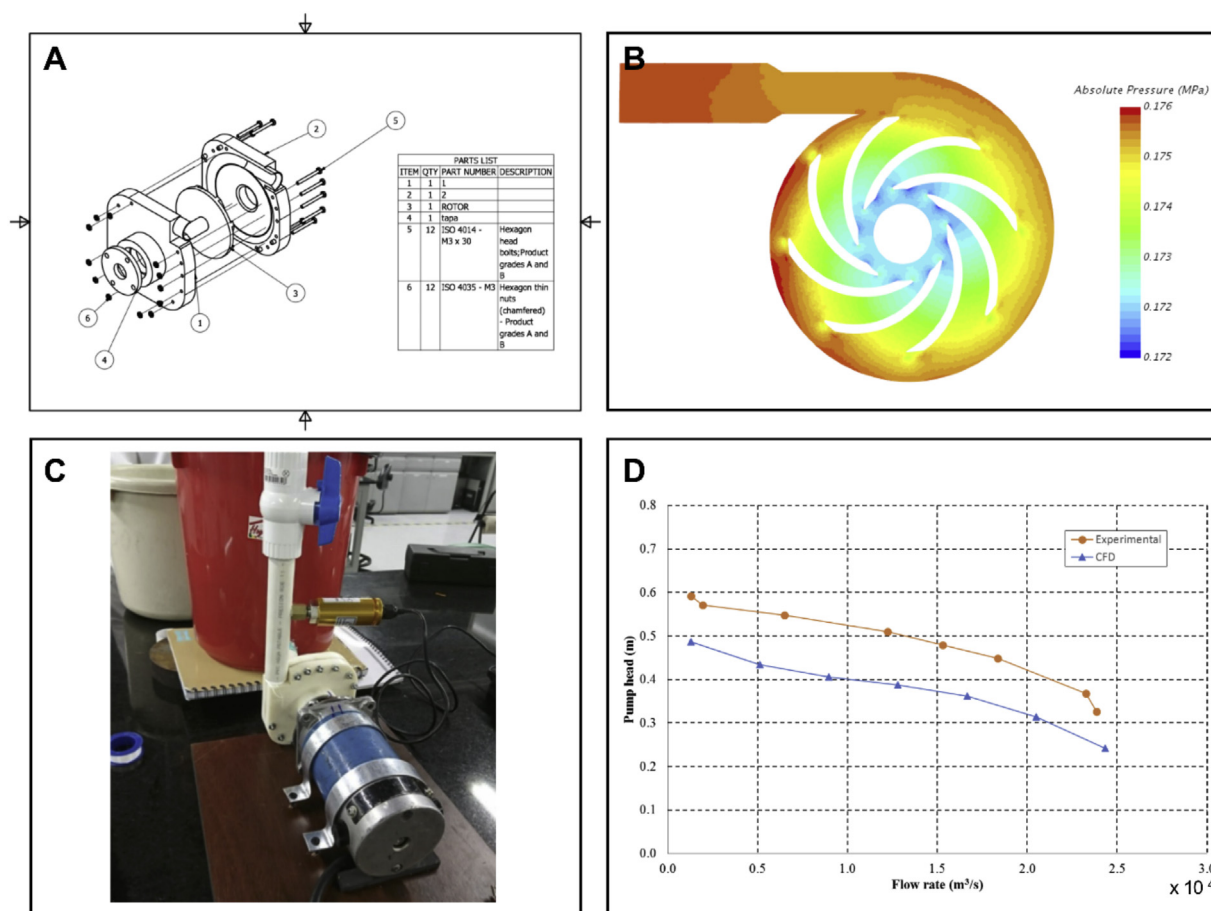


Fig. 5. Joint course project results. A. The centrifugal pump was designed in Autodesk Inventor®. B. CFD simulations were carried out to visualize and understand the phenomena taking place in the pump casing and impeller. C. Setup used for the students to test the performance of their 3D printed centrifugal pump. D. Comparison of the curves obtained using the experimental setup and the CFD simulations.

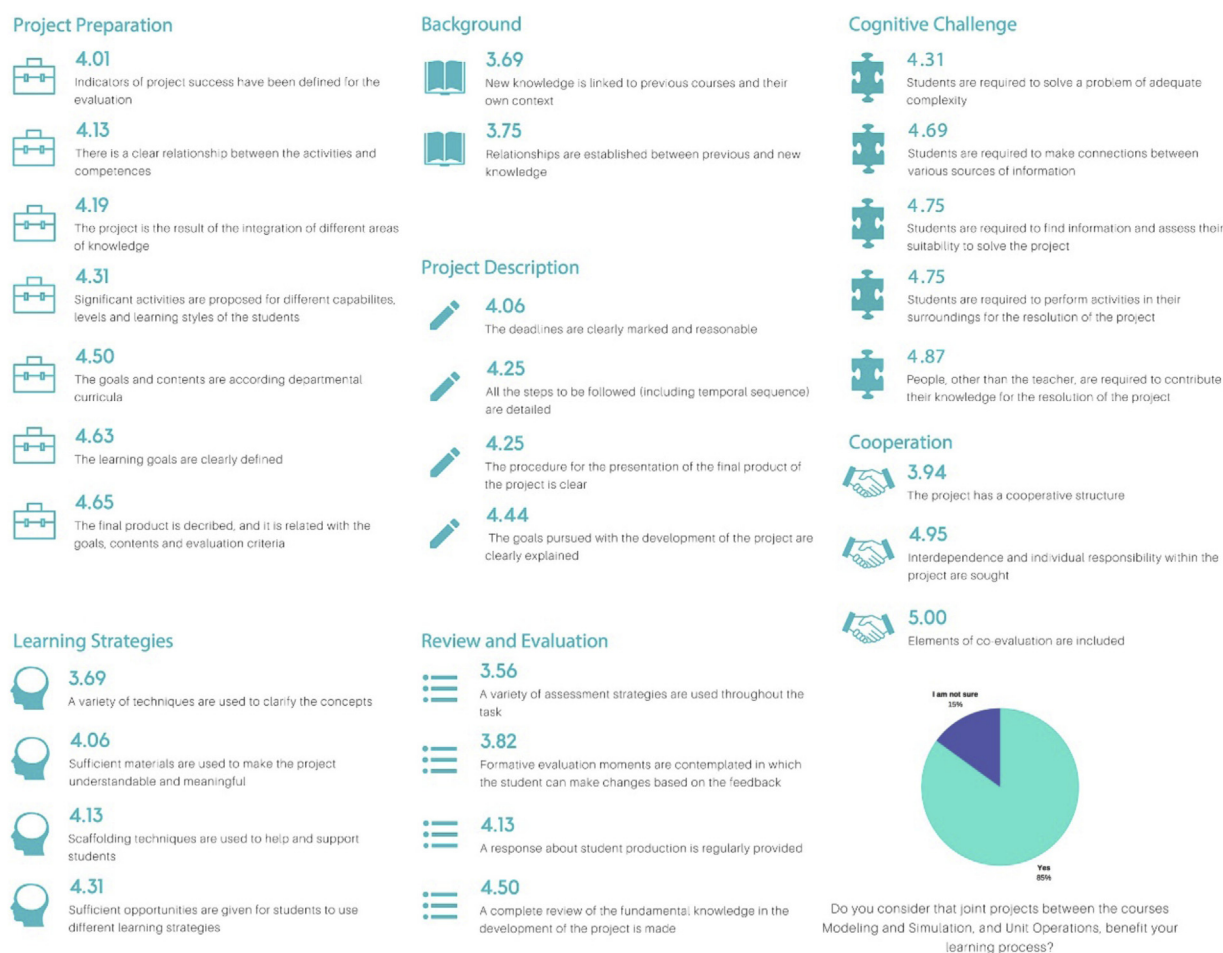


Fig. 6. Results of survey taken by students who completed the joint project (N = 15). All scores are on a 5 point scale where 5 is strongly agree and 1 is strongly disagree.

As for the results obtained on the student surveys (Fig. 6), it can be seen that the lowest scoring components on average was *Background*, with an arithmetic average score of 3.72 of 5.00, and *Review and Evaluation*, with an average of 4.00. On the other hand, the highest scoring aspects on average evaluated were the *Cognitive Challenge*, with an average grade of 4.67, and the *Cooperation* with an average score of 4.63. The other components considered have average scores ranging between 4.00 and 4.30. It should be noted that 21 students were enrolled in UnOps and 76 in SIM. 15 students were enrolled in both and completed the joint project described in this study. It is these 15 students that were surveyed to give the results in Fig. 6. The response rate for the 15 students was 100%.

The aforementioned values demonstrate the positive reaction of the students towards the project proposed and the results obtained. Additionally, they show a good acceptance of the methodology considered for the development of the project. This is clearly seen in Fig. 5, where 85% of the students answered positively when asked whether a PBL joint project contributed to their academic formation, and the remaining 15% answered dubitatively. There was no negative response at all towards the implementation of the joint project between UnOps and SIM. However, this 15% may be due to the methodology adopted by some of the students during the development of the project, in which specific individual tasks were divided among the members of each group. This may narrow the scope of the project and cause students to see the project as single and more menial tasks, rather than PBL. The introduction of the project helped both relate abstract theoretical concepts with an

example of a system widely used in the industry and make connection between different courses and their topics (UnOps and SIM). In doing so, the project fulfills the purpose of the LCT explained before, since it helps ground abstract ideas and avoids the compartmentalization of the topics of different courses.

The main strengths to be highlighted from the surveys performed correspond to the *Project Preparation* component. Specifically, the items related with the well-defined learning goals of the project, their relationship with the departmental curriculum and the close relationship of the final product and results obtained with the goals and contents of the project and of the courses as a whole. These criteria score over 4.50 out of 5.00, which shows that the project is consistent with the knowledge that is expected to be acquired from both courses: UnOps and SIM. Moreover, criteria such as integration of different areas of knowledge within the project, along with detailed definition of the step-by-step process, and the final goal of the project received high scores (over 4.20). However, the integration aspect has a lower score compared to the *Project Preparation* components, probably due to the considerable amount of material from both courses that is not included within the framework of the project. In order to address this issue, new transversal projects are being considered to guarantee a greater inclusion of the topics.

Furthermore, aspects related with the *Cognitive Challenge*, such as the requirement of new information, connections between various sources of information and different sources other than the professor for the development of the project are highly valued,

with scores above 4.70. In 2006, Federau describes a model to prepare appropriate problems for PBL in engineering (Federau, 2006). This model contemplates two axis, Assignment Freedom and Active Drive. Assignment Freedom considers how open a question is, and Active Drive how motivated a student will be to acquire the knowledge necessary to solve the problem. Good PBL problems should consider high degree of Assessment Freedom, promoting the students to deepen their knowledge further from the course program. This result shows that the complexity of the project encouraged the students to find and apply additional information to that acquired solely on the courses.

The main weaknesses and areas of improvement identified from the results rely primarily on the *Background* aspect, specifically related with the linking between the new knowledge acquired and the information from previous courses and own previous knowledge. Students in average did not connect or link clearly the knowledge being transmitted and applied on UnOps and SIM with other previous courses from the curriculum. This may be related to the fact that most of the contents of both courses contain new approaches that only depend partially on very basic or fundamental concepts from other subjects. Therefore, the PBL is more focused on learning and applying new concepts obtained from both courses, rather than reinforcing or linking previous fundamental bases seen in other subjects. This can be easily handled in the future by performing regular meetings with each group on the design phase, to ensure all the theoretical background is fully understood.

In addition, with the previously considered aspect, the variety of learning techniques and assessment strategies used throughout the project also obtained low scores, under 3.70. Students in average considered that the project and the courses did not offer them a diversity of learning strategies to reinforce and clarify concepts needed to carry out the task. Even though the project considered two completely separated learning environments (experimental and computational), the activities considered are not adequate enough to strongly reinforce the concepts being evaluated and applied on the development of the project. Additionally, given the considerable number of topics covered in both courses, the project's scope was not broad enough to apply and reinforce all of these concepts. As mentioned previously, this issue is being addressed through the design of transversal projects to be carried out throughout the subject. With these projects it can support even more the concept of LCT by applying it not only to turbomachinery but to other topics that are studied in both courses.

4. Conclusions

The PBL methodology gave the students a different approach to two of their major courses (UnOps and SIM) in junior year. The proposed joint project had the ability to reduce the workload of one of the most time-consuming year of the chemical engineering program. This joint project provided the students with various challenges: linking all the previous knowledge with the new acquired in the two courses, solving the driving question and creating a tangible product. The PBL methodology provided the students with new tools in managing ChEn problems.

The implementation of LCT could benefit the students as well as the PBL methodology. These two can give the students good tools for a transversal learning and understanding how concepts could be apply in real situations. Additionally, it can generate an active drive in the students to deepen their knowledge, beyond what is taught in the courses, and do a thorough research before making decisions in their projects. This drive to make well-informed and conscious decisions is a skill that would benefit their professional and personal future.

The results of the survey indicated that there are still topics, learning techniques, and assessment strategies that should be address with transversal projects to clarify the past knowledge and the new one. Nevertheless, 85% of the students answered positively on the benefits that the PBL joint project contributed to their academic formation. The idea is to reinforce the learning techniques with additional transversal projects. By complementing the joint project, the students will probably have a better and more well-rounded learning experience on all course-related topics.

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